Saurabh Nishad

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My LinkedIn Profile

Embedded Architect | Internet of Things | Product Developer | Speaker/Trainer

Profile

With around 8 years of experience collaborating with semiconductor companies, startups, and government/semi-government organizations, I am eagerly seeking an opportunity to contribute to organizational growth and development. My expertise spans Embedded Architecture, Internet of Things, Product Development, and Speaker/Training roles, making me a versatile professional with a passion for leveraging technology for impactful solutions.

EXPERIENCE

Sr. Application Engineer

STMicroelectronics Pvt. Ltd — *Bangalore*

STMicroelectronics is a Franco-Italian multinational electronics and semiconductors manufacturer.

April 2022 – Present

Roles and Responsibilities:

- Leading the development of Cellular and Cloud Connectivity Solutions, specifically utilizing AWS IoT Services, for STMicroelectronics in the APAC Region.
- Driving Reference Design Development to expedite startups and customers' go-to-market strategies.
- Providing hands-on support to customers in the design and development of their products.
- Delivering training sessions to STMicroelectronics' customers, ensuring comprehensive understanding and successful implementation.

Application Engineer

STMicroelectronics Pvt. Ltd. (3rd Party payroll "G4S Solution") — *Noida*

JANUARY 2020 – March 2022

Roles and Responsibilities:

- Contributed to Application/Middleware Development, extending ST's MCU/MPU/SoC functionality portfolio.
- Offered crucial support to customers in the design and development phases of their products.

Senior Embedded Software Engineer

Shellios Technolabs — *Noida NCR*

Shellios is a tech startup – It Is Born Out of The Need for A Revolution In Age-Old Helmet Industry.

FEBRUARY 2019 - JANUARY 2020

Roles and Responsibilities:

- Led a multidisciplinary team in the end-to-end design and development of embedded systems for Shellios's smart helmet product suite.

TECH STACK

Programming Languages

Language	Level
C (Embedded + System)	Expert
C++	Intermediate
C#	Intermediate
Python/Bash Scripting	Intermediate

Microcontrollers/Microprocessor Used for Projects

- ARM Cortex-M/A {Stm32 Microcontrollers and Microprocessor}
- STM8 Series of Controller
- Qualcomm Snapdragon 820
- AVR [Atmega328p, Atmega2560]
- Espressif [esp32, esp8266]
- 8051

Components Used

- LDO – (LD1117)
- LSM9DS1/LSM6DS0X (MEMS Sensor)
- M41T0 (RTC)
- M24SR (NFC EEPROM)
- VL53L1X (ToF Sensor)

Middleware Integration with ARM MCU

- LwIP (IP Suite)
- ST BLE Stack (Bluetooth)
- LoRAWAN Stack (LoRA Stack)
- TouchGFX (Graphic Library)
- USB Stack
- FatFS

Consultant and Freelancer

(Smarttech3DP/Sennet Control Gmbh.)

Sennet Control Gmbh: It's a German based company manufacturing electronic devices of sensor technology and measurement technology, as well as the operation of a calibration laboratory for physical and electrochemical measurements.

Smarttech3Dp is US based Company working in manufacturing of Electronic device and IoT Products of consumer market.

JANUARY 2018 - DECEMBER 2018

Roles and Responsibilities:

- Provided technical support for product development, focusing on Firmware Development and IoT Connectivity Solutions.

Embedded Firmware Engineer

Intello Transpo Pvt. Ltd. — *Noida NCR*

JANUARY 2017 - DECEMBER 2017

Roles and Responsibilities:

- Designed a complete Bootloader for OTA with Quectel M95 (GSM Module) and developed firmware for the VTS Project.

Embedded Trainee Engineer

STmicroelectronics Pvt. Ltd. — *Greater Noida*

JANUARY 2016 - DECEMBER 2016

Roles and Responsibilities:

- Contributed to firmware development and demo application design.
- Designed an NFC-based IR Thermometer for measuring body temperature.

EDUCATION

Instrumentation and Control Engineering

Galgotias College of Engineering and Technology — *B-TECH*

MAY 2012 - APRIL 2016

Achievements

- Presented on **"IoT on Edge Device Taking to Cloud"** at an **AICTE-sponsored ATAL, FDP on "Internet of Things (IOT)" in 2021.**
- Published a research paper on **"Low-Cost Smartphone Controlled Wireless UAV"** in the IEEE International Conference on **"Sustainable Energy, Electronics & CoMputing Systems 2018."**
- Holds a provisional patent for **"An Automated Cell Counting Device for Early Detection of Leukemia and other Blood Disorders."**
- Delivered a tutorial talk on **"Industrial IoT: Its Vertical Applications in Real-Time Systems"** at the IEEE International Conference on **"Sustainable Energy, Electronics & CoMputing Systems 2018."**
- Participated in Imagine Cup and completed a project on Microsoft Technology as a **"Microsoft Student Partner,"** specializing in Microsoft IoT technologies.

Embedded Operating System

- FreeRTOS (RTOS)
- Azure-RTOS
- BuildRoot (Embedded Linux)
- Yocto (Embedded Linux)
- Zephyr (RTOS)

Operating Systems

- Linux [Ubuntu/Debian]
- Windows

IDEs

- STM32CubeIDE/EWARM/Keil – C Development
- Visual Studio Code
- Android Studio – For Android Prototyping
- QT – Linux Application Prototyping

Version Control System

- Git
- SVN

Productivity Tools

- JIRA
- Doxygen
- Latex
- Confluence

Miscellaneous

- Design-Pattern (MVP/MVVM)
- Linux System programming
- Linux Kernel build process
- Linux Kernel programming
- Data Structures
- Arduino based rapid prototyping
- KiCad/ Altium Schematic design
- Embedded Hardware Design
- Electronics Production
- Oscilloscopes
- Power Supply Analysis
- Node-Red
- AWS – (IoT-Core/Greengrass/EC2/Lambda)